

**YEAR 10 SCIENCE EXTENSION
PHYSICS
ASSIGNMENT 2**

NAME: _____

MARK: $\frac{\quad}{45}$

1. A plane is flying at an altitude of 10,000 m at $7.90 \times 10^2 \text{ kmh}^{-1}$ W. There is a strong wind blowing from the south at $1.50 \times 10^2 \text{ kmh}^{-1}$. Determine the plane's velocity over the ground. Include a vector diagram.

(4)

2. An athlete runs $6.00 \times 10^2 \text{ m}$ E in 95.0 s, then $8.00 \times 10^2 \text{ m}$ SE in 2.00 minutes.
- (a) Calculate her displacement.

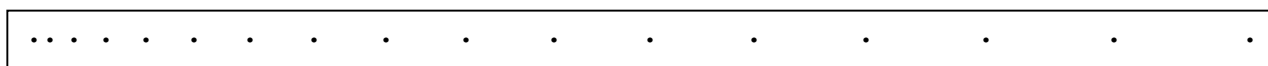
(3)

(b) Determine her average speed.

(c) What is the athlete's average velocity? (2)

(3)

3. In recording the motion of a toy car down an inclined track, a student produced the following tape using a ticker timer operating at 50 dots per second. Analyse the tape and determine the acceleration of the car. (Use the diagram below.)



(4)

4. A car undergoes the following motion between three sets of traffic lights A, B and C.

The car accelerates uniformly from rest at light A and reaches 22.0 ms^{-1} after 8.00 s . It maintains this velocity for 20.0 s before slowing uniformly as it approaches lights B. The lights turn green so the car, having reduced its velocity to 10.0 ms^{-1} in 4.00 s , accelerates uniformly for 10.0 s to reach 25.0 ms^{-1} . It maintains this velocity for 5.00 s before uniformly decelerating to a stop in 5.00 s at lights C.

- (a) Draw a velocity-time graph for the motion. (The scales only need to show the important data and need not run evenly.)

(4)

- (b) Using the graph (or otherwise), calculate the size of the decelerations that occurred during the motion.

(4)

- (c) How far apart are lights A and C?

(4)

5. A downhill skier of mass 80.0 kg started from rest on top of a mountain and accelerated smoothly down a uniform slope. After moving 4.00×10^2 m his velocity was $1.80 \times 10^2 \text{ kmh}^{-1}$.

- (a) Calculate the acceleration experienced by the skier.

(3)

- (b) How long did it take for him to travel this far?

(3)

- (c) Would the skier actually have the velocity mentioned in the question? Explain your answer.

(2)

6. A boy throws a stone upwards at 15.0 ms^{-1} from the edge of a cliff 25.0 m high. It just misses the edge on the way down and falls to the bottom of the cliff.

- (a) Calculate the maximum height reached by the stone above the ground.

(3)

- (b) With what velocity does the stone strike the ground?

(3)

- (c) How long is the stone in the air?

(3)